Journal of Hydrology 663 (2025) 134190

Contents lists available at ScienceDirect

Journal of Hydrology

journal homepage: www.elsevier.com/locate/jhydrol

Research papers

Predicting the isotopic composition of precipitation in China via interpretable machine learning techniques

Jingming Liu [a,b], Jianli Ding [c,*], Jinjie Wang [a,b,*], Jie Zou [a,b], Zipeng Zhang [a,b], Qingling Bao [a,b], Xiao Wang [a,b], Xiangyu Ge [a,b]

[a] College of Geographical and Remote Sciences, Xinjiang University, Urumqi 830017, China
[b] Xinjiang Key Laboratory of Oasis Ecology, Xinjiang University, Urumqi 830017, China
[c] Xinjiang Institute of Technology, Aksu, China

ARTICLE INFO

This manuscript was handled by Huaming Guo, Editor-in-Chief, with the assistance of Philippe Negrel, Associate Editor

Keywords:
Stable isotopes
Machine learning
SHapley Additive exPlanations
China

ABSTRACT

Stable isotopes δ^2H and δ^{18}O, which form the composition of water molecules, serve as intrinsic tracers that illuminate the historical information regarding the evolution of the water cycle. The analysis of hydrogen and oxygen isotopes in precipitation proves valuable in discerning the mechanisms governing water cycle dynamics across diverse regions. Despite the existence of a multitude of observation networks and studies worldwide, the analysis of different temporal and regional scales in China remains constrained by the availability of data. We employed two interpretable machine learning models and utilize accessible climate variables and regional classification variables to predict precipitation isotopes. Furthermore, we evaluated the accuracy of these predictions using various metrics. Notably, random forest regression exhibits superior performance in predicting precipitation isotopes in China, particularly with regard to its sensitivity to continuous and categorical variables. The SHapley Additive exPlanations effectively explain the role of variables in model predictions, with temperature and precipitation variables having a stronger influence. The interpretable nature of machine learning facilitates the uncomplicated and convenient prediction of precipitation isotope time series. Ultimately, this approach can be employed to explore interannual and regional variations within hydrogen and oxygen isotope systems.

1. Introduction

Stable isotope techniques, renowned for their accuracy, are widely utilized in the study of the water cycle, climate change, and geological processes (Gu et al., 2011). The isotopes δ^2H and δ^{18}O are commonly employed in investigating precipitation. By analyzing the proportions of these isotopes, valuable insights can be gained regarding the source, formation mechanisms, and spatial variations of precipitation (Dansgaard, 1964). International interest in studying precipitation isotopes emerged in the 1980s. The Global Network of Isotopes in Precipitation (GNIP), established in 1961, has collected extensive long-term data on this subject. Several countries have also formed their own networks to observe precipitation isotopes, such as the Austrian Network of Isotopes in Precipitation (ANIP, Kralik et al., 2003), the Canadian Network of Isotopes in Precipitation (CNIP, Fritz et al., 1987), and the United States Network of Isotopes in Precipitation (USNIP, Lynch et al., 1995). Unfortunately, the majority of GNIP stations in China ceased monitoring activities in the early 2000s. In response, the China Network of Isotopes in Precipitation (CHNIP) was established in 2004 (Song et al., 2007), which also initiated the observation of isotopes in precipitation on the Tibetan Plateau (Yu et al., 2016). Nonetheless, due to the vast geographical expanse of China and significant variations in environmental geology distribution, there is a dearth of time-series data in comparison to the densely-distributed stations in Europe.

Precipitation isotope prediction models have reached a mature stage, with spatial interpolation methods, such as regression and geostatistical coupling, gradually becoming mainstream. These regression relationships and known parameter grid data enable the prediction of precipitation isotope ratios. Notable representative products in this field include the online isotopes in precipitation calculator (OIPC, Bowen and Wilkinson (2002)), and the regionalized cluster-based water isotope prediction (RCWIP, Terzer et al., 2013). Various dynamic simulation methods, such as CAM2 to CAM5 and isoGSM to isoGSM2, have been developed and may be extended to different types of climate models

* Corresponding authors.
E-mail addresses: watarid@xju.edu.cn (J. Ding), skytian552@163.com (J. Wang).

https://doi.org/10.1016/j.jhydrol.2025.134190
Received 24 May 2024; Received in revised form 10 August 2025; Accepted 28 August 2025
Available online 31 August 2025
0022-1694/© 2025 Published by Elsevier B.V.

(Wang et al., 2023). Despite advancements, the collection of measured data remains fragmented, even in densely populated European regions, and there are still inadequate applications of isotopes. Recognizing this gap, Nelson et al. developed a framework called Piso AI. By employing machine learning, Piso AI utilizes climate variables and spatial location data to calculate isotopic monthly time series and improve the prediction accuracy of European precipitation isotope models. The model enables predictions at any location in Europe during the available climate variables period spanning the last 70 years (1950–2019) (Nelson et al., 2021). Erdelyi et al. employed geostatistics and machine learning methods to predict the spatial distribution of stable isotopes in European precipitation. They also evaluated the performance of various interpolation techniques in predicting such spatial distribution (Ait Brahim et al., 2016; Erdélyi et al., 2023). In a similar vein, Heydarizad et al. employed artificial neural networks, stepwise regression, wavelet coherence, and integrated machine learning methods to predict precipitation isotopes in Tehran (Heydarizad et al., 2023). Notably, machine learning algorithms eliminate the need for stationarity assumptions and address concerns related to anisotropy or fitting variance (Bataille et al., 2018; Hengl et al., 2018).

China exhibits a vast regional expanse and encompasses diverse climate types, yet suffers from limited and unevenly distributed weather stations. Numerous studies have examined the spatial distribution and characteristic variations of precipitation isotopes in China. Liu et al. conducted an analysis on the range of variation in stable hydrogen and oxygen isotopes in atmospheric precipitation in China, while Zhang et al. examined the hydrological processes associated with precipitation isotopes in the same country (Liu et al., 2008). Chen et al. analyzed the spatiotemporal distribution characteristics of stable isotopes in precipitation in China, along with their influencing factors (Chen et al., 2010a; Chen et al., 2010b). Wang et al. investigated the spatial–temporal distribution of information entropy of precipitation isotopes in China and traced their transport processes within moisture. The findings collectively demonstrate significant climatic, continental, and altitudinal effects on precipitation isotopes (Wang et al., 2022a; Wang et al., 2022b). Furthermore, a substantial number of studies have explored the correlation between precipitation isotopes and climate factors across different regions of China (e.g. (Zhang and Yao (1994); Liu et al. (2009); Zhang et al. (2015); Duan et al. (2020); Chang et al. (2021); Chen et al. (2022); Zhang et al. (2023)).

In terms of research methods, Liu et al. employed the BW model in combination with spatial interpolation to simulate the spatial distribution of oxygen stable isotopes in precipitation in China (Liu et al., 2009). Chen et al. produced a precipitation oxygen isopleth map for China over a span of fifty years using data fusion and bias correction (Chen et al., 2023). Building upon regional fuzzy clustering, Wang et al. generated diagrams depicting the spatial and seasonal variations of isotopes in precipitation across China (Wang et al., 2022a; Wang et al., 2022b), offering robust methods for predicting isotopes in regional precipitation. However, there is a dearth of analysis cases in the context of artificial intelligence.

The primary objective of this study is twofold. Firstly, a range of well-established machine learning algorithms, such as Random Forest (RF) and eXtreme Gradient Boosting (XGBoost), will be applied to forecast precipitation isotopes in China. The efficacy of the developed models will be assessed through accuracy evaluation and optimal parameter selection to identify the most suitable algorithm and model. Secondly, this study seeks to investigate the mechanisms by which climate variables and regional parameters influence precipitation isotopes in China. The processes employed in the models will be globally and locally explained based on the SHAP (SHapley Additive exPlanation) theory.

2. Materials and methods

2.1. Used data and preprocessing

Given the monthly resolution of the GNIP observations, the dataset offers a long-term and continuous time series. Monthly δ^2H and δ^{18}O values for precipitation from 1980 to 2022 were retrieved from the GNIP database (IAEA/WMO, 1992). From these records, we additionally calculated the deuterium excess (d-excess) values, which were used as response variables in subsequent modeling. In total, 1698 observations from multiple GNIP sites were compiled (Fig. 1). In addition to the CNIP station records, we incorporated the monthly-scale precipitation δ^{18}O database for the Tibetan Plateau, which contains 698 observations (Yao et al., 2013). Furthermore, we supplemented the dataset with 3523 additional records compiled from published literature for China, obtained from the C-Isoscape Database (https://cisoscape.wz.dlszysy.net.cn) (Wang et al., 2022b). This integration of multi-source datasets substantially enhanced the spatial and temporal coverage of precipitation isotope observations across China, thereby providing a robust basis for subsequent predictive modeling. The study incorporated three main datasets: Dataset 1, comprising a complete set of paired δ^2H and δ^{18}O measurements; Dataset 2, containing the same variables without seasonal stratification; and Dataset classified by distinct geographic zones (Fig. 1). Prior to analysis, the original data underwent preprocessing to address missing values and ensure temporal consistency. Considering the presence of extreme climate events, no outliers were removed during data processing.

In terms of predictor variables, we employed latitude, longitude, and elevation measured at GNIP stations, together with site-specific observational data from each region (Appendix Fig. S1). We further incorporated spatiotemporal categorical predictors, including region, month, and season, but excluded year to avoid potential collinearity with long-term climate trends. Beyond the climate variables provided by GNIP stations, we integrated a suite of additional climatic predictors from the National Centers for Environmental Prediction (NCEP; Kalnay et al., 1996) and the National Center for Atmospheric Research (NCAR) reanalysis datasets. Moreover, teleconnection indices were compiled as time-series variables from climate prediction centers, including the El Niño–Southern Oscillation (ENSO), Western Pacific Subtropical High Area Index (WPSHAI), Western Pacific Subtropical High Intensity Index (WPSHII), Western Pacific Subtropical High Western Ridge Point Index (WPSTHWRPI), West Pacific Pattern (WPP), and the West Pacific 850 mb Trade Wind Index (WP850mbTWI) (Barnston et al., 1987; Wallace and Gutzler, 1981).

Collectively, these efforts yielded a total of 41 predictor variables, encompassing geographic, climatic, teleconnection, and categorical components, thereby ensuring comprehensive coverage of both local and large-scale atmospheric controls on precipitation isotopes.

2.2. Methods applied

2.2.1. Machine learning model

Two machine learning algorithms that demonstrate strong performance in handling complex data structures were chosen considering the attributes of the available data. These algorithms have yielded favorable outcomes in predicting precipitation isotopes in the European region. The random forest regression prediction method is an ensemble learning approach grounded in decision trees. It encompasses the construction and combination of multiple decision trees to enable regression prediction. The construction of decision trees incorporates randomness, including the random selection of features and samples for training. This randomized process aids in mitigating the risk of model overfitting and effectively capturing intricate relationships and patterns within the data (Biau and Scornet, 2016; Breiman, 2001). The eXtreme Gradient Boosting (XGBoost) regression prediction method is based on the gradient boosting algorithm. It progressively enhances prediction

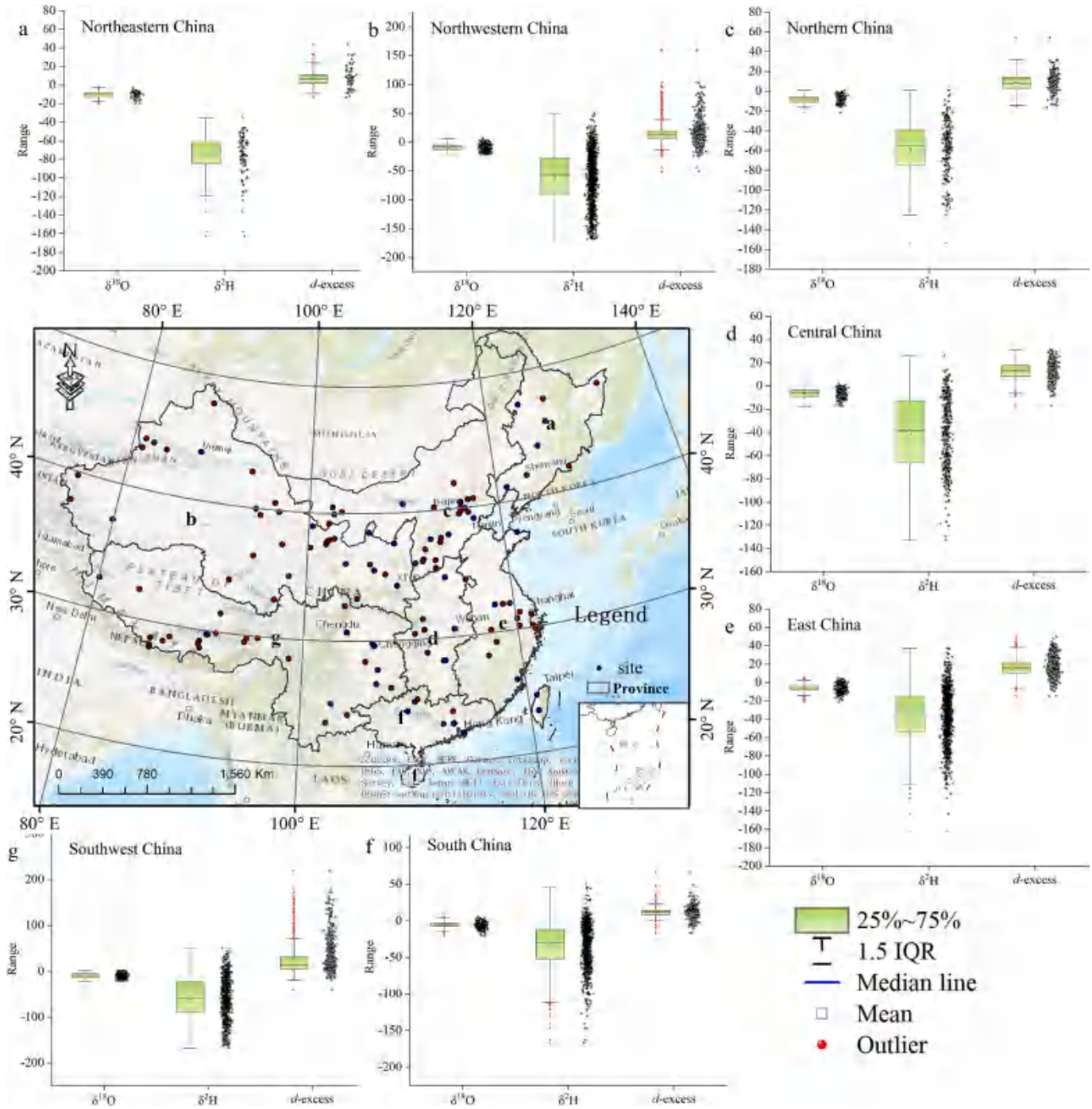

Fig. 1. Study area scope, regional units, and site location distribution.

performance by iteratively training multiple decision tree models. Notably, this approach exhibits strong proficiency in addressing regression problems characterized by continuous target variables (Chen and Guestrin, 2016).

2.2.2. Hyperparameter optimization

Grid Search is a method that tries all combinations of hyperparameters within a given search space and ultimately searches for the optimal combination of hyperparameters. The specific steps are as follows. Assuming we have k hyperparameters, and for each hyperparameter, there are n_i candidate values, the total number of combinations is N. Cross-validation (CV(θ)) is employed to assess the model's performance for each hyperparameter combination. The optimal hyperparameter combination, denoted as θ^*, is selected based on the performance metric obtained from cross-validation, with mean squared error (MSE) being the assumed metric (Kuhn, et al., 2020).

$$N = \Pi_{i=1}^{k} n_i \tag{1}$$

$$\mathrm{CV}(\theta) = \frac{1}{\kappa}\sum_{j=1}^{\kappa} MSE(\theta, D_j) \tag{2}$$

$$\theta^* = \arg{*}min_{\theta\in\Theta} CV(\theta) \tag{3}$$

where "D_j" represents the validation dataset in the j-th fold, "$MSE(\theta, D_j)$" represents the mean squared error when using the hyperparameter combination "θ" on the validation set "D_j".

Bayesian Optimization, on the other hand, is a probabilistic model-based optimization method that selects the next hyperparameter combination to be evaluated by constructing a probabilistic model of the objective function. A Gaussian process model is constructed by using prior probability distributions and existing data. The Gaussian process assumes that the target function $f(\theta)$ follows a multivariate Gaussian distribution in the space of hyperparameters. For a given set of observed points $\{(\theta_i, y_i)\}_{i=1}^{n}$, the predictive distribution is represented by the mean function $\mu(\theta)$ and the covariance function $k(\theta, \theta')$.

$$f(\theta) \ \mathscr{GP}(\mu(\theta), k(\theta, \theta')) \tag{4}$$

$$\mu(\theta_*) = k(\theta_*, \Theta)^T K^{-1} y \tag{5}$$

$$\sigma^2(\theta_*) = k(\theta_*, \theta_*) - k(\theta_*, \Theta)^T K^{-1} k(\theta_*, \Theta) \tag{6}$$

where "Θ" represents all evaluated hyperparameter combinations, "y" represents the corresponding observation values, "K" is the covariance matrix, and $k(\theta_*, \Theta)$ is the covariance vector between a new hyperparameter combination "θ_*" and the evaluated hyperparameter combinations.

Expected Improvement (EI), a collection function, is defined and optimized to determine the subsequent hyperparameter combination for evaluation. The process is repeated until the hyperparameter combination that yields the best performance is obtained.

$$\alpha_{EI}(\theta) = E[max(0, f(\theta^+) - f(\theta))] \tag{7}$$

where "θ^+" represents the current known best hyperparameter combination, and $f(\theta)$ represents the distribution predicted by the surrogate model.

2.2.3. Shapley Additive exPlanations

The SHapley Additive exPlanations (SHAP) method is employed to explain predictions made by machine learning models. It is based on the Shapley value derived from cooperative game theory. The SHAP value is computed by making predictions with various feature values and assessing the influence of each specific feature value on the prediction outcome. This facilitates the explanation of the predictive significance of individual features within the model and the contributions made by different feature values to the overall prediction outcome (Soetaert, 2024).

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i \tag{8}$$

where "ϕ_0" represents the baseline value (usually the average predicted value of all samples), and "ϕ_i" represents the contribution of the first feature.

2.3. Model performance evaluation

The R-squared (R^2), also referred to as the coefficient of determination, is a statistical measure utilized to evaluate the quality of fit of a regression model. It quantifies the extent to which the model elucidates the variability in the observed data, specifically reflecting the proportion of variance that can be accounted for by the model.

$$R^2 = 1 - \frac{SS_{rse}}{SS_{tot}} \tag{9}$$

where "*SSres*" refers to the sum of squared residuals, which quantifies the sum of squared differences between the actual observed values and the predicted values of the model. On the other hand, "*SStot*" represents the total sum of squares, which signifies the sum of squared differences between the actual observed values and the mean value of the observed data.

The Root Mean Square Error (*RMSE*), also referred to as the root mean square error, is a statistical measure employed to evaluate the predictive errors of a regression model. It quantifies the average deviation between the predicted values generated by the model and the observed values.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n} (\delta_i - \widehat{\delta}_i)^2} \tag{10}$$

where "n" denotes the number of samples, " δ_i" signifies the i-th observed value, "$\widehat{\delta}_i$" represents the model-predicted value corresponding to the i-th observation.

Mean Absolute Error (*MAE*) is a statistical metric utilized for evaluating the prediction error of regression models. It quantifies the average absolute deviation between the predicted values of the model and the observed values.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |\delta_i - \widehat{\delta}_i| \tag{11}$$

where "n" denotes the number of samples, " δ_i" signifies the i-th observed value, "$\widehat{\delta}_i$" represents the model-predicted value corresponding to the i-th observation,

The Concordance Correlation Coefficient (*CCC)* is a statistical metric used to evaluate the agreement between predicted and observed values. Unlike correlation coefficients that measure only the strength of a linear relationship, the *CCC* simultaneously assesses both precision (the degree of correlation) and accuracy (the closeness to the 1:1 line), providing a more comprehensive measure of model performance.

$$CCC = \frac{2\rho\sigma_x\sigma_y}{\sigma_x^2 + \sigma_y^2 + (\mu_x - \mu_y)^2} \tag{12}$$

where ρ denotes the Pearson correlation coefficient between the observed (x) and predicted (y) values; $\sigma_x^2 and \sigma_y^2$ represent the variances of observed and predicted values, respectively; $\mu_x and \mu_y$ are the means of the observed and predicted values, respectively.

3. Results

3.1. Model process

In this study, the entire dataset was randomly divided into a training set and a test set at a ratio of 0.75. Hyperparameter tuning was performed on the training set using cross-validation, and the optimal hyperparameters were subsequently applied to train the final model on the training set. The test set was then used for model performance evaluation, comparative analysis, and subsequent interpretability assessment (Appendix Fig. S2). To mitigate bias introduced by random partitioning, five-fold cross-validation was further applied within the training set for model training and performance evaluation.

For hyperparameter optimization, two global search strategies were employed: grid search and Bayesian optimization. Cross-validation was performed on the training set with stratified sampling to ensure representative evaluation across different training–validation splits, and the results of the two optimization methods were compared. The random forest model was optimized using grid search, whereas the XGBoost model adopted Bayesian optimization. The results are presented in Appendix Fig. S3.

Once the optimal hyperparameters were determined, they were applied to the complete training set to build the final model, which was then used to generate predictions for the test set in order to evaluate and compare model performance, as well as to produce subsequent interpretability analyses. Following the procedure illustrated in Fig. 2, the optimized models were further applied to Dataset 2. For optimal hyperparameter combination, evaluation metrics were computed through cross-validation, and the corresponding tables and figures were generated (Appendix Tables S1).

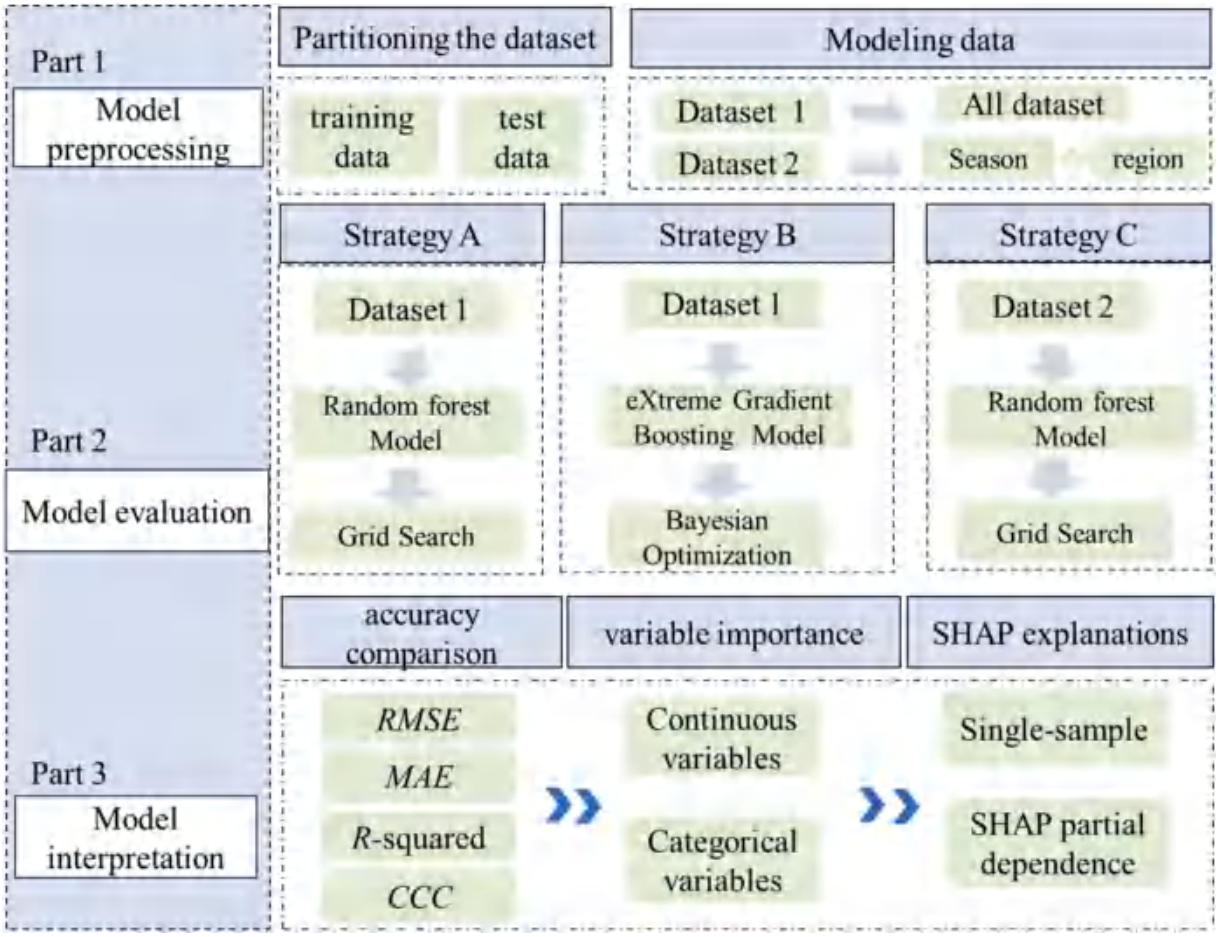

Fig. 2. Conceptual diagram of the model prediction analysis framework.

Significant regional differences in δ^2H, δ^{18}O, and d-excess were observed across the study area. The quartile plots (Fig. 1) indicate that δ^2H exhibits the greatest dispersion, followed by d-excess, whereas δ^{18}O shows the most concentrated distribution. Outliers are relatively scarce for δ^2H and δ^{18}O, but more prevalent for d-excess, largely because d-excess is a derived variable calculated from δ^2H and δ^{18}O.

The accompanying figures illustrate the dataset partitioning results, showing that both the training and test sets fall within the range of the complete dataset, thereby avoiding data bias and confirming the rationality of the partitioning (Appendix Figs. S4 and S5). Scatterplots comparing model-predicted and observed values (Fig. 3) reveal that the two models exhibit comparable accuracy, with the δ^{18}O model achieving the best predictive performance, followed by the δ^2H model, and the *d*-excess model performing the poorest. The suboptimal performance of the *d*-excess model is primarily attributed to the extrapolative nature of the variable itself, a finding consistent with conclusions from European precipitation prediction studies (Nelson et al., 2021). In terms of model comparison, the Random Forest (RF) model slightly outperformed the XGBoost model in predicting δ^2H, δ^{18}O, and *d*-excess.

When comparing the present study's Piso.AI model (*RMSE* values of 1.68 ‰, 13 ‰, and 3.93 ‰ for δ^{18}O, δ^2H, and *d*-excess, respectively) with the Tehran XGBoost model (*RMSE* values of 1.97 ‰ and 12.54 ‰ for δ^{18}O and δ^2H, respectively), the prediction accuracy in our study area is slightly lower, likely due to the denser observational network in Europe. Nevertheless, the results are broadly comparable to those obtained using the fuzzy clustering method at the regional scale in China (R^2 of 0.67 for δ^{18}O prediction). When applying the CNN-fusion approach, 80.4 % of the stations in the Chinese precipitation oxygen isopleth map achieved simulated values with an *RMSE* of less than 3 ‰ over a half-century timescale, which is comparable to the *RMSE* accuracy of the present model. Notably, in precipitation isotope prediction for Tehran, the XGBoost model outperformed the RF model, likely reflecting differences in regional data characteristics. Considering multiple accuracy metrics, the proposed model demonstrates strong generalizability. Moreover, compared with fuzzy clustering and CNN-fusion methods, the model developed in this study is more computationally efficient.

3.2. Prediction of factor importance

For δ^2H, both the Random Forest and XGBoost models (Fig. 4 A and B) identify Month (seasonality) and Altitude as the most important predictors. These are followed by near-surface climate factors such as dew point temperature and minimum air temperature, along with geographic variables like latitude, all of which rank highly in feature importance. The two modeling approaches show broadly consistent rankings for δ^2H, indicating that seasonal cycles and elevation-driven effects dominate its variability. Similarly, for δ^{18}O, the models (Fig. 4 C and D) agree that Altitude is the single most influential feature, closely followed by average station pressure and Month. This emphasizes that elevation and seasonal factors strongly govern δ^{18}O values. Additionally, meteorological variables – for example, average dew point temperature and measures of precipitation (e.g., NCEP reanalysis precipitation and local station precipitation) – emerge as important contributors in δ^{18}O predictions. These climate factors likely capture moisture source and "amount effect" influences on δ^{18}O, complementing the dominant altitude and seasonal controls. Furthermore, the role of categorical variables was examined (Appendix, Fig. S6). In the δ^{18}O

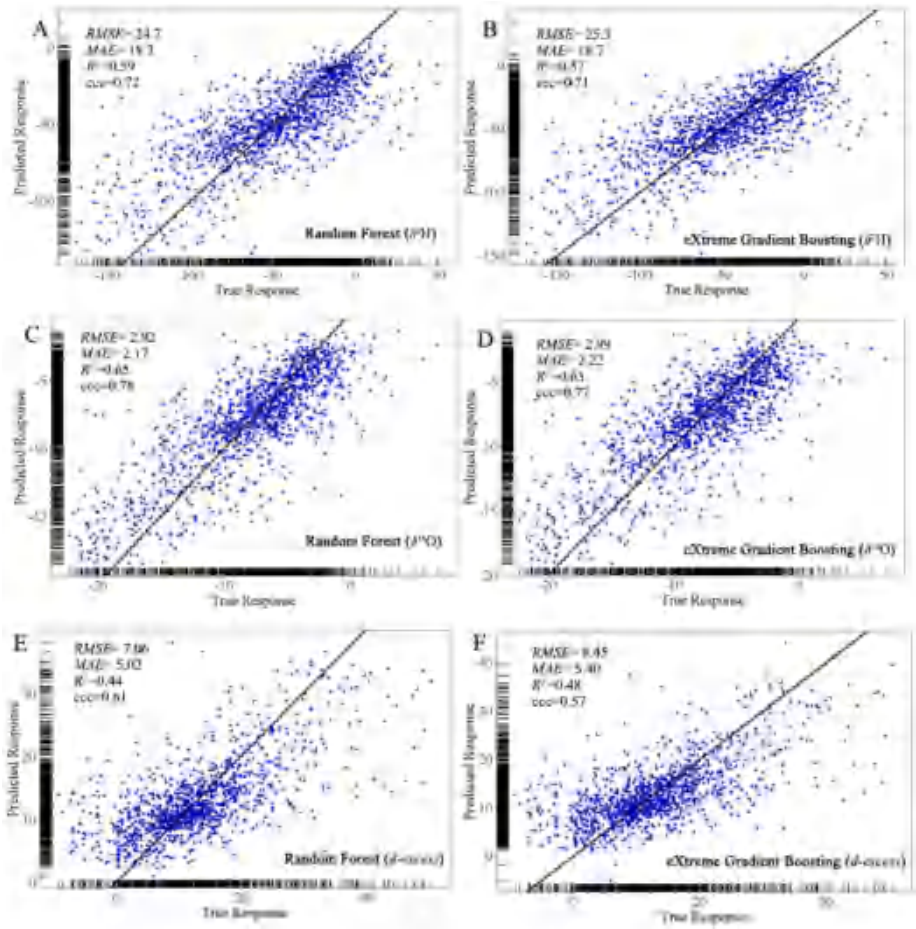

Fig. 3. Figures A, C, E display the fitting accuracy of the random forest model for δ^2H, δ^{18}O, and d-excess. Figures B, D, F show the fitting accuracy of the eXteme Gradient Boosting model for δ^2H, δ^{18}O, and d-excess.

modeling in particular, these categorical predictors show a notable yet modest influence. For instance, grouping factors like climate or region exhibit consistent contributions to δ^{18}O predictions, but they remain secondary compared to the leading continuous variables. This suggests that while categorical distinctions (such as regional climate zones or seasonal groupings) do affect δ^{18}O, their effects are largely captured through the major continuous predictors discussed above.

3.3. Accuracy of simulations across different seasons and regions

3.3.1. Accuracy of seasonal simulations

The Random Forest model predictions for δ^{18}O and δ^2H exhibited marked seasonal differences in fitting accuracy. (Fig. S7 and S8). For δ^{18}O, the model performed best in winter ($R^2 = 0.76$, $CCC = 0.85$, $RMSE = 2.90$), indicating that δ^{18}O variations in winter can be effectively captured by the model (Fig. 7). In contrast, the lowest accuracy was observed in summer, where scatter points were more dispersed, reflecting the high variability of monsoonal precipitation processes and the influence of multiple water vapor sources, which increased prediction uncertainty (Rhoujjati et al., 2023). For δ^2H, the highest accuracy was also achieved in winter ($R^2 = 0.67$, $CCC = 0.81$, $RMSE = 27.0$). Although the *RMSE* was relatively large due to the wider numerical range of δ^2H, the predicted trends were highly consistent with the observations. Spring and autumn showed comparable accuracy, both presenting relatively stable fitting trends.

Overall, δ^{18}O generally showed higher predictive accuracy than δ^2H across all seasons, with particularly pronounced differences in autumn and winter, indicating that δ^{18}O responds more stably to large-scale temperature fields and atmospheric circulation. In contrast, δ^2H is more sensitive to local evaporation and mixed vapor sources, resulting in greater prediction uncertainty. Both isotopes exhibited a consistent seasonal pattern of highest accuracy in winter and lowest in summer, underscoring the high instability of monsoonal precipitation and its constraining effect on isotope simulation accuracy.

3.3.2. Accuracy of region simulations

The predictive accuracy of δ^{18}O is consistently higher than that of δ^2H across all regions. Both isotopes exhibit relatively high correlations and concordance (high R^2 and *CCC)* in Southwest, Northwest, and Central China, whereas prediction accuracy in Northeastern China is limited, likely due to insufficient sample size. Spatial variations in *RMSE* and *MAE* indicate that prediction errors in the extreme value ranges—particularly for δ2H in the Northwest and Southwest—are markedly higher than in the mid-range, suggesting that the model's capacity to capture extreme isotopic signals requires further improvement (Table 1).

4. Discussion

4.1. SHAP partial dependence

The SHAP partial dependence plots (Fig. 8) reveal pronounced nonlinear relationships and threshold effects in the Random Forest model's predictions of precipitation δ^{18}O. Altitude emerges as a dominant control factor: SHAP values decrease sharply with increasing elevation, showing strongly negative contributions at high altitudes

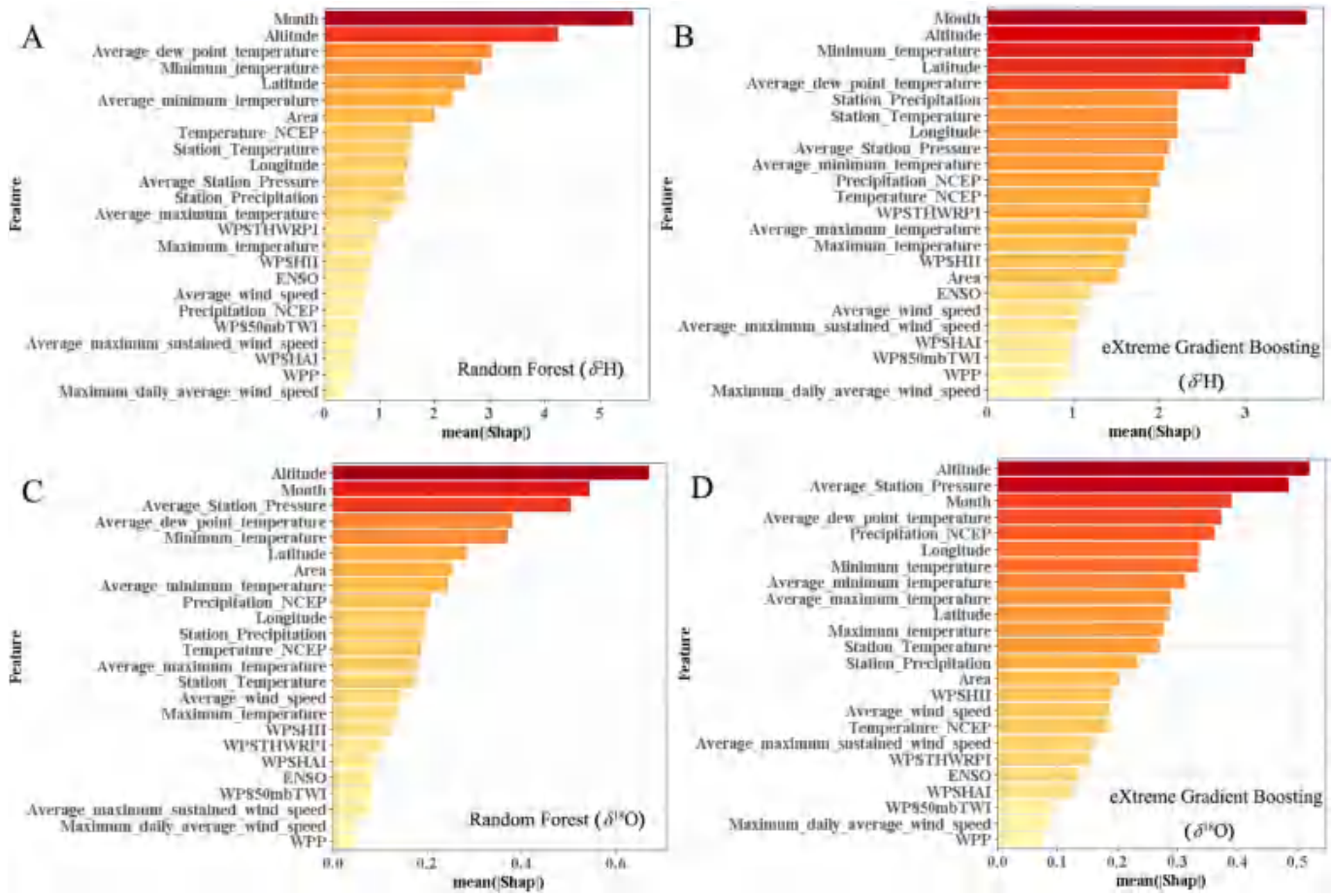

Fig. 4. The ranking of climate variable importance. The importance of climate variables for δ^2H and δ^{18}O in the random forest model is shown in Figure A and C. The importance of climate variables for δ^2H and δ^{18}O in the eXtreme Gradient Boosting model is shown in Figure B and D.

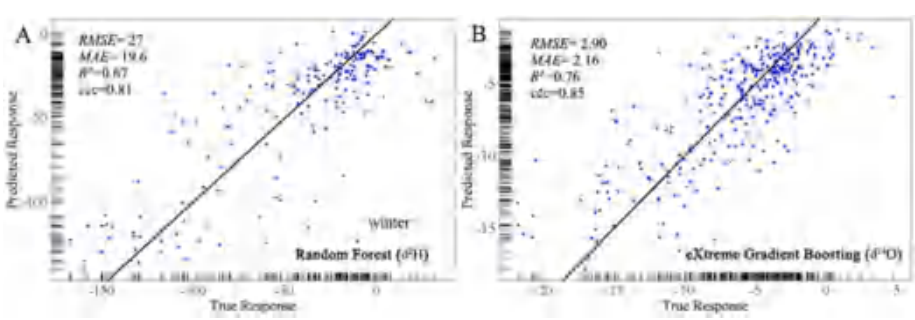

Fig. 7. The prediction results are analyzed for winter. Figures A demonstrates the prediction accuracy of δ^2H, Figures B demonstrates the prediction accuracy of δ^{18}O.

Table 1
Prediction accuracy of models in different regions.

Area	δ^2H (‰)				δ^{18}O (‰)			
	R-squared	*RMSE*	*MAE*	*CCC*	*R-squared*	*RMSE*	*MAE*	*CCC*
Northeastern China	0.29	18.4	13.8	0.45	0.493	2.69	2.30	0.63
Northwestern China	0.58	29.8	22.4	0.72	0.608	3.57	2.77	0.74
Northern China	0.44	21.4	16.4	0.56	0.514	2.84	2.04	0.60
Central China	0.581	19.3	15.1	0.71	0.597	2.32	1.75	0.68
East China	0.51	20	15.2	0.64	0.429	2.55	1.91	0.65
South China	0.60	19.1	13	0.71	0.478	2.53	1.73	0.63
Southwest China	0.43	34.3	25.7	0.58	0.675	3.44	2.70	0.77

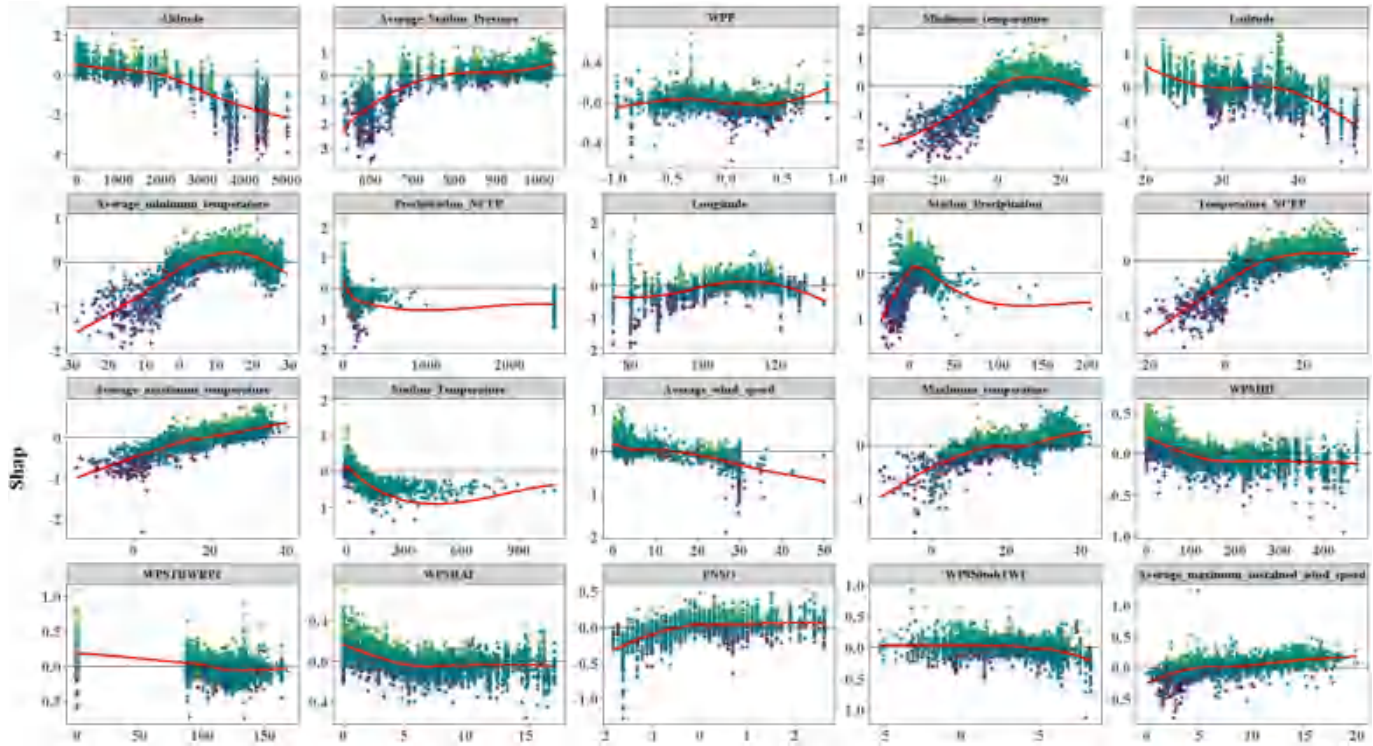

Fig. 8. Variable feature SHAP dependency plots for $\delta^{18}O$ in different regions.

(Zeng et al., 2021). Features representing precipitation amount (including station observations and NCEP reanalysis data) exhibit clear threshold behavior: once a certain rainfall amount is exceeded, SHAP values decline substantially (Zhu et al., 2022a; Zhu et al., 2022b). Intense rainfall events shift $\delta^{18}O$ towards more negative values. During periods dominated by subtropical high pressure—characterized by dry, subsiding air—rainfall processes are weaker, resulting in relatively heavier isotopic ratios (Yang, 2020). In contrast, low-pressure systems promote rainfall processes and isotopic depletion. Temperature exerts a significant but complex nonlinear influence on $\delta^{18}O$. Both minimum and maximum temperature features indicate that under extremely cold conditions, SHAP values are negative whereas mild temperatures are associated with positive SHAP contributions ($\delta^{18}O$ enrichment). Latitude shows a gradual decline in SHAP values with increasing latitude, suggesting slightly more negative $\delta^{18}O$ in higher-latitude regions. Although the teleconnection indices have smaller amplitudes, their effects are consistent with known influences on moisture transport (Zhan et al., 2023). Overall, the SHAP analysis demonstrates that the Random

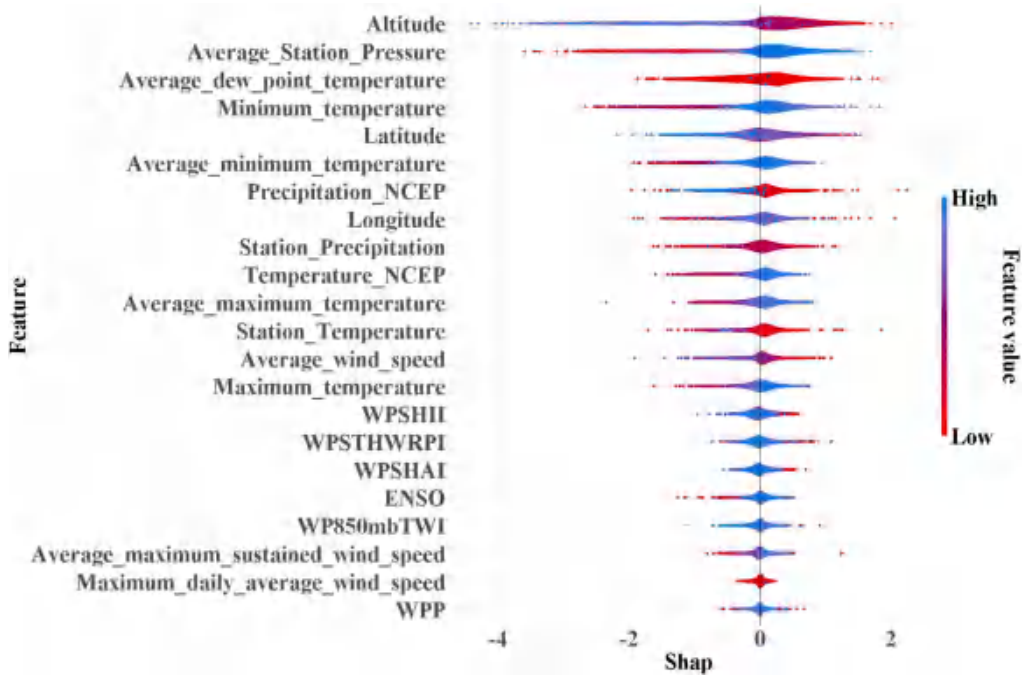

Fig. 9. The SHAP partial dependence plots of continuous variables for $\delta^{18}O$.

Forest model has captured plausible isotope–climate relationships. The discussion of these SHAP-derived relationships further reinforces our process-based interpretation of the isotope data.

4.2. Model interpretation based on different types of variables

4.2.1. Continuous variable

The SHAP of variables align with the distribution of variable importance. In the SHAP summary plot (Fig. 9), each feature's impact is represented by colored points, revealing clear patterns across the feature gradients. Altitude is a highly important feature in the prediction of both δ2H and δ1^{8}O, with high-altitude conditions corresponding to markedly negative SHAP values, indicating that the model predicts more depleted isotope values under such conditions. high precipitation amounts correspond to strongly negative SHAP contributions for δ^2H/δ^{18}O, indicating that intense rainfall events drive these isotope values lower. This aligns with the well-known "amount effect," whereby heavy monsoonal precipitation causes isotopic depletion through enhanced rainout and convective recycling. In contrast, low precipitation yields positive SHAP values, meaning drier conditions lead to higher isotope ratios. Likewise, temperature related variables exhibit coherent behavior: higher temperatures and dew points (often occurring in summer monsoon months) push δ^2H/δ^{18}O toward more negative values, reflecting the influence of warm, moisture-rich air masses from tropical oceans that produce isotopically depleted rain. By contrast, lower temperatures and drier air (characteristic of winter and continental air masses) are associated with positive SHAP effects, corresponding to more isotopically enriched precipitation. The high and low values of the teleconnection index do affect the prediction, albeit insignificantly. The impact of other feature variables on the prediction can be observed in (Appendix, Fig. S9).

4.2.2. Categorical variables

Both isotopic models exhibit clear seasonal patterns in their SHAP value distributions, reflecting the well-known seasonal cycle of precipitation isotopes (Hu and Wang, 2015; Wang et al., 2012; Wu et al., 2015). In the δ^2H models (Fig. 10), certain months consistently have positive SHAP values, while others have negative SHAP values. Specifically, the drier or less rainy months tend to show positive median SHAP contributions, whereas the peak wet-season months (mid-year in monsoon climates) yield strongly negative median SHAP values (Zhao, 2017; Zhu et al., 2022a; Zhu et al., 2022b). Seasonal SHAP boxplots also reveal that some transition months exhibit asymmetrical impacts. The Random Forest and XGBoost results for δ^2H agree on the overall seasonal pattern and sign of effects, though XGBoost tends to show a slightly wider spread for certain month categories.

The SHAP boxplots for categorical region variables reveal how location-based factors affect the model outputs for δ^2H and δ^{18}O, and they highlight some differences in spatial influence between the two isotopes. Distinct regions show clearly different SHAP value distributions, indicating that the model adjusts predictions up or down depending on the site's region. Some regions consistently yield negative

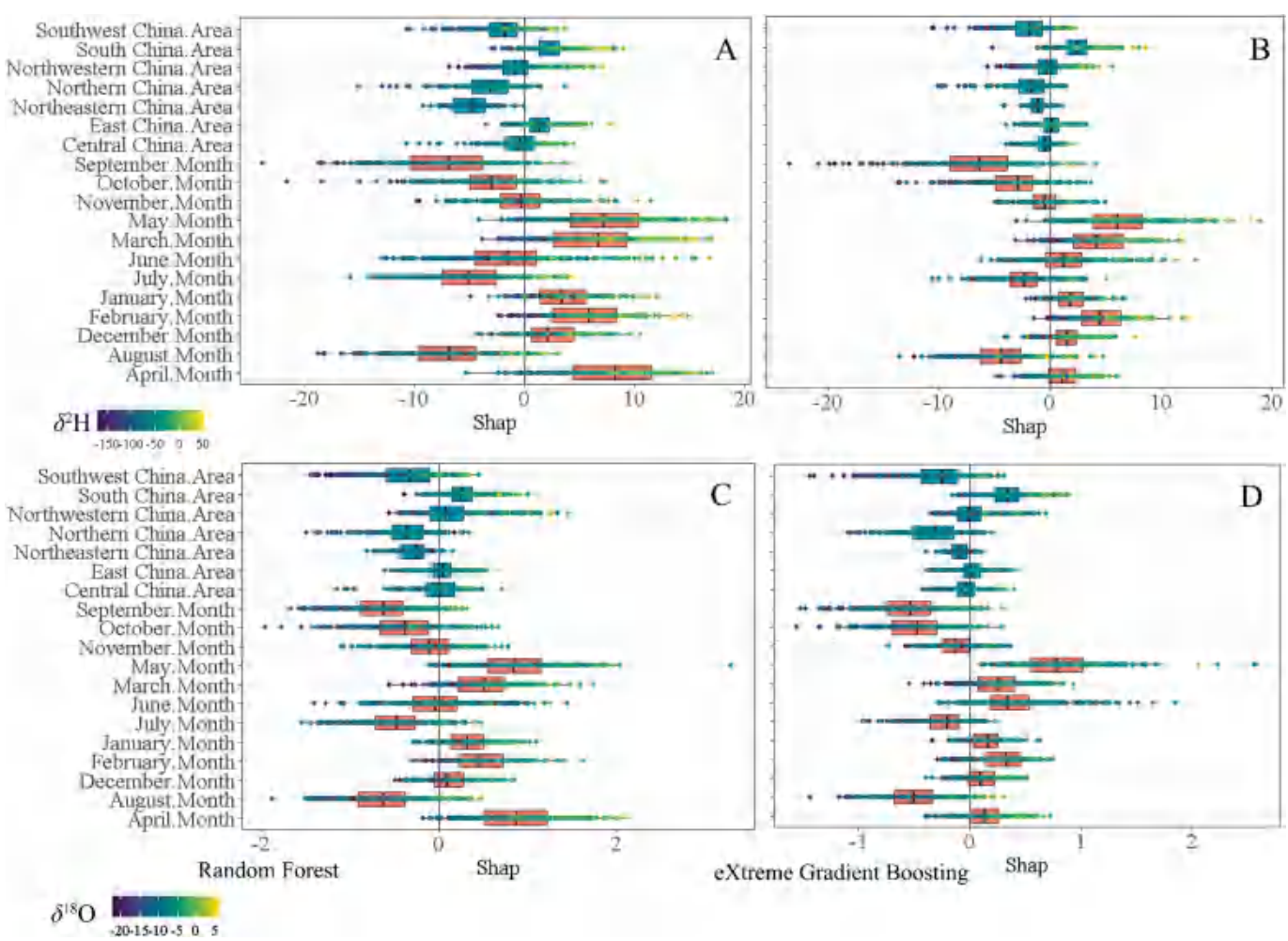

Fig. 10. The SHAP plots of categorical variables, Figure A and B shows the distribution of SHAP values for δ^2H, Figure C and D shows the distribution of SHAP values for δ^{18}O.

SHAP values, meaning those locations tend to lower the predicted δ2H relative to the overall average. This suggests those regions have more depleted hydrogen isotope values, likely due to climatic factors like high rainfall or cooler temperatures (Duan et al.,2016; Chen et al., 2010). By contrast, other regions have positive SHAP contributions for δ^2H, indicating a regional tendency toward higher δ^2H. Such positive regional effects could correspond to areas with drier climate or significant evaporation influence, where precipitation is isotopically heavier.

4.3. Single-sample analysis

The single-sample SHAP analysis for δ^{18}O predictions (Fig. 11) reveals pronounced seasonal differences in how key environmental features influence precipitation isotope values. In spring, the low δ^{18}O values at this sample site are primarily driven by longitude and low-temperature indicators (dew point temperature and minimum temperature), indicating a strong sensitivity of spring δ^{18}O to temperature effects. In summer, the negative influence of WPS850mbTWI suggests that intense moisture transport delivers oceanic-sourced precipitation, during which δ^{18}O is substantially depleted, resulting in more negative values. The positive effect of altitude may be associated with the prevailing atmospheric circulation background, leading to relatively higher δ^{18}O. In autumn, δ^{18}O enrichment is predominantly controlled by geographic location (low latitude and coastal high-pressure zones) and the transport of warm, humid air masses. The combination of high atmospheric pressure and low latitude points to subsidence warming under the influence of the subtropical high, coupled with marine moisture input, which markedly elevates δ^{18}O during this season (Jin et al., 2019; Li et al., 2022). In winter, δ^{18}O depletion is mainly driven by the combined effects of higher latitude, low temperatures, low dew point, and limited precipitation, consistent with the control of cold, dry air masses originating from the Siberian High (Wang et al., 2015; Wang et al., 2019). During long-distance transport, cold air masses undergo progressive isotopic depletion, and the weak re-evaporation in winter almost completely suppresses δ^{18}O enrichment (Xing et al., 2018). Single-sample SHAP values based on the Random Forest model. (see Fig. 10).

4.4. SHAP partial dependency for different regions

Across all regions, the major climatic drivers of precipitation δ^{18}O exhibit pronounced nonlinearity and marked spatial heterogeneity (Fig. 12 and Appendix Fig. S12). In Northeast China, the latitudinal effect is consistent with variations in temperature and water vapor transport pathways: increasing latitude corresponds to lower temperatures, longer moisture transport distances, and enhanced condensation, thereby intensifying isotopic fractionation and yielding more negative δ^{18}O values (Fig. 12a). Furthermore, temperature indices also play a dominant role (Appendix Fig. S12a) (Wang et al., 2018; Zhang et al., 2021). In the arid Northwest, δ^{18}O exhibits a strong nonlinear dependence on temperature variables, with temperature serving as a key controlling factor (Appendix Fig. S11b). The SHAP partial dependence plot (Fig. 12b) shows that at very low mean minimum temperatures, SHAP values are strongly negative, indicating substantial depletion of δ^{18}O in precipitation under extremely cold conditions. This pattern reflects the classic "temperature effect" in the inland Northwest: higher temperatures generally correspond to higher δ^{18}O values. In particular, during the hot and dry summer, precipitation is scarce and often occurs as localized convective storms, with raindrops experiencing substantial evaporation during descent, enriching the remaining precipitation in

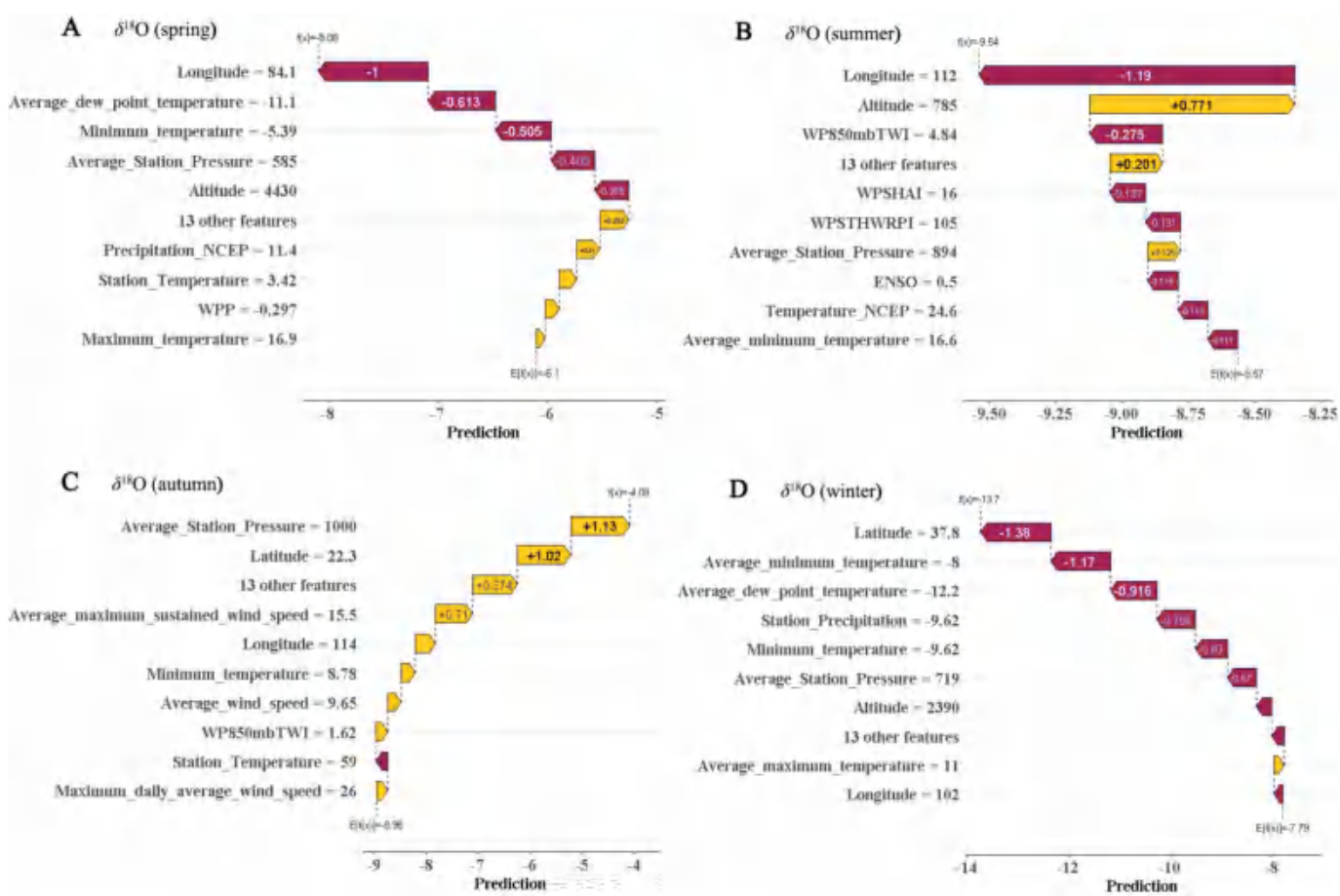

Fig. 11. Single-sample SHAP plots. Figures A to D demonstrate the SHAP values of δ^{18}O.

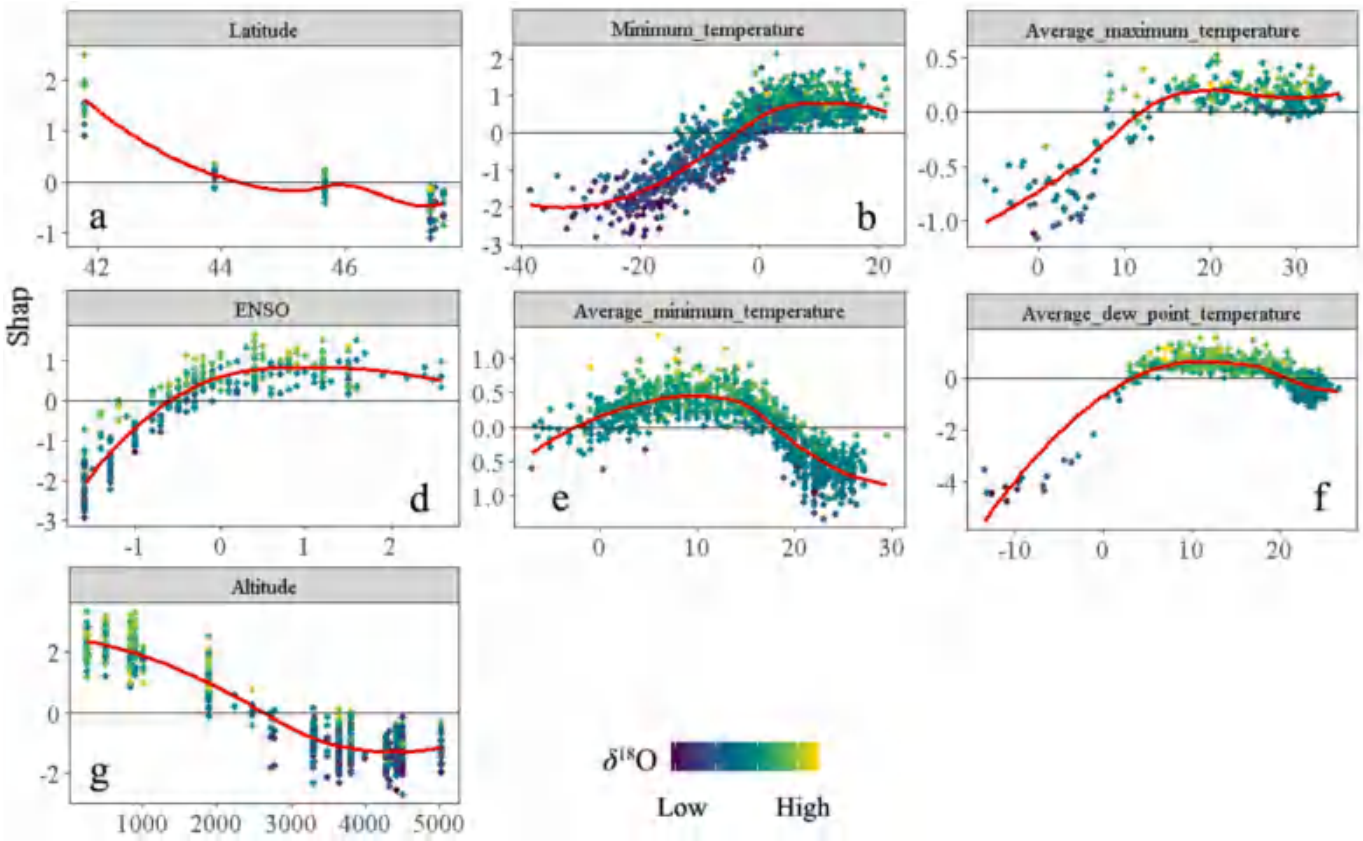

Fig. 12. SHAP partial dependence plots of the most important variables of $\delta^{18}O$ in each region, where a ≃ g corresponds to different regions.

heavy isotopes. This mechanism highlights the combined control of temperature and re-evaporation effects on $\delta^{18}O$ under continental climate conditions (Tian et al., 2007; Cao et al., 2021; Rao et al., 2021).

In North China, $\delta^{18}O$ follows the temperature effect: as temperature increases, condensation fractionation during precipitation weakens, leading to enriched $\delta^{18}O$ values. Under low-temperature conditions, stronger condensation fractionation causes marked $\delta^{18}O$ depletion, reflected in strongly negative SHAP contributions (Fig. 12c) (Zhao et al., 2018; Zhang et al., 2015; Zhang, 2020). In Central China, the ENSO index exerts a significant teleconnection influence on precipitation $\delta^{18}O$, manifesting as a nonlinear SHAP dependence (Fig. 12d). When the ENSO index is positive, SHAP values shift to positive, indicating $\delta^{18}O$ enrichment. The SHAP curve shows an inflection point around ENSO = 0, with near-zero influence under neutral conditions. Under strong El Niño or La Niña conditions, the effect tends to saturate, suggesting that the marginal influence of ENSO on $\delta^{18}O$ diminishes during extreme events (Cui et al., 2025). Such behavior is analogous to the nonlinear responses observed for other climatic variables in this region and is consistent with findings from previous studies (Zhao et al., 2009; Wang et al., 2018) (Appendix Fig. S12d). In the East China monsoon region, $\delta^{18}O$ shows a complex coupling with temperature and precipitation processes, being mainly influenced by these two factors (Fig. 12d and Appendix Fig. S12e). The relationship between mean minimum temperature and $\delta^{18}O$ in East China is strongly dependent on the regional climatic background: a threshold of approximately 15–20 °C marks a turning point from positive to negative impacts (Liu et al., 2010; Shi et al., 2021). In South China, a typical humid monsoon climate region, mean dew point temperature exerts a particularly strong nonlinear effect on precipitation $\delta^{18}O$. This asymmetric "dry–wet" pattern reveals that processes such as re-evaporation influence precipitation isotopic composition in markedly different ways under varying humidity conditions (Appendix Fig. S12f). Under extremely low dew points and dry air, raindrops undergo substantial re-evaporation during descent, enriching the remaining precipitation in heavy isotopes (Wu et al., 2013). This mechanism underscores the regulatory role of re-evaporation on $\delta^{18}O$ in southern precipitation (Meng and Liu, 2010; Wu et al., 2015; Chen et al., 2015; Zhao et al., 2015). In the Southwest Plateau and mountainous areas, the SHAP dependence plot (Fig. 12g) reveals a clear negative relationship with a nonlinear weakening trend. As elevation increases from low altitudes to about 2000 m, SHAP values decrease from near zero to negative, consistent with the classic "altitude effect" (Yao et al., 2013). At elevations above 3000 m, the slope of the SHAP curve flattens and the negative values approach saturation. This suggests that once air masses are lifted to high elevations, most water vapor has already condensed at lower altitudes, heavily depleting heavy isotopes; the residual vapor is extremely depleted, so further uplift yields minimal additional depletion, and precipitation $\delta^{18}O$ approaches a lower limit (Zhang et al., 2006; Shi et al., 2020; Liu et al., 2014).

5. Conclusions and outlook

This study integrates machine learning models with SHAP-based interpretability analysis, employing multiple parameter selection strategies and comprehensive accuracy metrics to investigate the climatic controls and spatiotemporal variability of $\delta^{18}O$ and $\delta^{2}H$ in precipitation across China. The results reveal that both isotopes exhibit consistent and physically interpretable patterns across multiple spatial and temporal scales.

Seasonally, the highest prediction accuracy for both isotopes occurs in winter, while the lowest is observed in summer. This seasonal contrast reflects the increased complexity of warm-season monsoonal precipitation, the diversity of moisture sources, and the high variability of convective activity. Regionally, $\delta^{18}O$ predictions outperform those of $\delta^{2}H$, indicating that $\delta^{18}O$ responds more stably to large-scale

temperature and circulation patterns. SHAP partial dependence analysis identifies altitude, precipitation, and temperature indices as the dominant predictors, all exhibiting strong nonlinear relationships and threshold effects. High-elevation regions and heavy precipitation events are associated with pronounced isotope depletion, consistent with the well-established altitude and precipitation amount effects. Temperature effects show seasonal asymmetry: cold conditions drive isotope depletion. Teleconnection indices exert secondary regulatory effects by modulating moisture transport pathways. These findings confirm that combining machine learning models with interpretability tools can effectively capture physically meaningful isotope–climate relationships, consistent with established process understanding. Future work should focus on providing a more intelligent and efficient predictive framework for precipitation isotope analysis.

6. Analytical tools

All machine learning, data analysis, and mapping were done using RStudio, version 1.2.5001, and R, version 3.6.0 (47). Required R packages included Tidymodels (Kuhn et al., 2020), fastshap (Greenwell, 2024), ggh4x (van den Brand, 2024), randomforest (Liaw and Wiener, 2002), xgboost (Geng et al., 2024).

CRediT authorship contribution statement

Jingming Liu: Writing – original draft. **Jianli Ding:** Conceptualization. **Jinjie Wang:** Data curation. **Jie Zou:** Resources. **Zipeng Zhang:** Validation. **Qingling Bao:** Conceptualization. **Xiao Wang:** Supervision. **Xiangyu Ge:** Methodology.

Funding

This research was supported by the Technology Innovation Team (Tianshan Innovation Team), Innovative Team for Efficient Utilization of Water Resources in Arid Regions (No. 2022TSYCTD0001). the Excellent Doctoral Innovation Project of Xinjiang University (XJU2024BS077). The effect of inter-basin watertransfer on the ecological effects of Xinjiang's Lake Ebinur (42171269).

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jianli Ding reports was provided by the Technology Innovation Team (Tianshan Innovation Team), Innovative Team for Efficient Uti-lization of Water Resources in Arid Regions(NO.2022TSYCTD0001). Jianli Ding reports was provided by the effect of inter-basin watertransfer on the ecological effects of Xinjiang's Lake Ebinur (42171269). Jingming Liu reports was provided by the the Excellent Doctoral Innovation Project of Xinjiang University (XJU2024BS077). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jhydrol.2025.134190.

Data availability

Publicly available datasets were analyzed in this study, which can be found at: GNIP database. https://www.iaea.org/water Climate variable data can be found at. https://www.cpc.ncep.noaa.gov. precipitation $\delta^{18}O$ database for the Tibetan Plateau can be found at: https://www.tpdc.ac.cn/zh-hans/data. The C-Isoscape Database (http://cisoscape.wz.dlszysy.net.cn).

References

Ait Brahim, Y., Bouchaou, L., Sifeddine, A., Khodri, M., Reichert, B., Cruz, F.W., 2016. Elucidating the climate and topographic controls on stable isotope composition of meteoric waters in Morocco, using station-based and spatially-interpolated data. J. Hydrol. 543, 305–315. https://doi.org/10.1016/j.jhydrol.2016.10.001.

Barnston, A.G., Barnston, R.E., Livezey, 1987. Classification, seasonality and persistence of low-frequency atmospheric circulation patterns. Weather Rev. 115, 1083–1126.

Bataille, C.P., et al., 2018. A bioavailable strontium isoscape for Western Europe: a machine learning approach. PLoS One 13 (5), e0197386. https://doi.org/10.1371/journal.pone.0197386.

Biau, G., Scornet, E., 2016. A random forest guided tour. TEST 25 (2), 197–227. https://doi.org/10.1007/s11749-016-0481-7.

Bowen, G.J., Wilkinson, B.H., 2002. Spatial distribution of d18O in meteoric precipitation. Geology 30, 315–318. https://doi.org/10.1130/0091-7613(2002)030,0315: SDOOIM.2.0.CO;2.

Breiman, L., 2001. Random forests. Mach. Learn. 45 (1), 5–32. https://doi.org/10.1023/A:1010933404324.

Cao, L., Shen, J.M., Nie, Z.L., et al., 2021. Stable isotope characteristics of precipitation and water vapor recycling in the Badain Jaran Desert. Earth Sci. 46 (08), 2973–2983. https://doi.org/10.3799/dqkx.2020.273.

Chang, X., Zhang, X. P., Dai, J. J., Liu, Z. I., 2021. Comparison of stable isotope effects under different time scales: taking Changsha as an example. Chinese Quaternary Sci., 41(41), 99-110, 10.11928/j.issn.1001-7410.2021.01.09.

Chen, F., Zhang, M., Wang, S., et al., 2015. Relationship between sub-cloud secondary evaporation and stable isotope in precipitation of Lanzhou and surrounding area. Quat. Int. 380 (381), 68–74. https://doi.org/10.1016/j.quaint.2014.12.051.

Chen, J., Zhang, X.J., et al., 2023. A century and a half Precipitation Oxygen Isoscape for China Generated using Data Fusion and Bias Correction. Sci. Data 10 (1). https://doi.org/10.1038/s41597-023-02095-1.

Chen, J.F., Cao, J.P., Huang, Y.P., 2010a. Stable hydrogen and oxygen isotopic composition of atmospheric precipitation and its influencing factors in the coastal area of Xiamen. J. Marine Sci. 28 (01), 11–17.

Chen, T., Guestrin, C., 2016. XGBoost: a scalable tree boosting system. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 785–794. https://doi.org/10.1145/2939672.2939785.

Chen, Y.N., Li, Z., Fang, G.H., 2022. Changes of key hydrological elements and research progress of water cycle in the Tianshan Mountains, Central Asia. Arid Land Geogr. 45 (1), 8. https://doi.org/10.12118/j.issn.1000-6060.2021.535.

Chen, Z.X., et al., 2010b. Distribution characters and its control factors of stable isotope in precipitation over China. Trans. Atmos. Sci. 33 (6), 13. https://doi.org/10.3969/j.issn.1674-7097.2010.06.004.

Cui, Y., Tian, L., Cai, Z., Wang, S., 2025. Spatially inhomogeneous response of precipitation δ18O in China to ENSO cycles. npj Clim. Atmos. Sci. 8 (1), 164. https://doi.org/10.1038/s41612-025-01057-1.

Dansgaard, W., 1964. Stable isotopes in precipitation. Tellus 16 (4), 436–468. https://doi.org/10.3402/tellusa.v16i4.8993.

Duan, L.Q., Liu, F.H., Su, Y.J., Kong, S.Q., 2020. The relationship between $\delta^{18}O$ in precipitation in Eastern China and the ENSO. Chinese J. Irrigat. Drainage 39 (9), 7.

Duan, W., Ruan, J., Luo, W., Li, T., Tian, L., Zeng, G., Tan, M., 2016. The transfer of seasonal isotopic variability between precipitation and drip water at eight caves in the monsoon regions of China. Geochim. Cosmochim. Acta 183, 250–266. https://doi.org/10.1016/j.gca.2016.03.037.

Erdélyi, D., Hatvani, I.G., Jeon, H., et al., 2023. Predicting spatial distribution of stable isotopes in precipitation by classical geostatistical- and machine learning methods. J. Hydrol. 617. https://doi.org/10.1016/j.jhydrol.2023.129129.

Fritz, P., Drimmie, R.J., Frape, S.K., et al., 1987. The isotopic composition of precipitation and groundwater in Canada. Atmos. Precipit. 539–550.

Geng, Y., Li, Y., Yuan. J., 2024. xgboost: Extreme Gradient Boosting_. R package version 1.7.7.1. https://CRAN.R-project.org/package=xgboost>.

Greenwell, B., 2024. fastshap: Fast Approximate Shapley Values_. R package version 0.1.1. https://CRAN.R-project.org/package=fastshap>.

Gu, W.Z. et al., 2021. Isotope hydrology, Beijing: Science Press, 2011.

Hengl, T., Nussbaum, M., Wright, M.N., Heuvelink, G.B., Gräler, B., 2018. Random forest as a generic framework for predictive modeling of spatial and spatio-temporal variables. PeerJ 6, e5518.

Heydarizad, M., Gimeno, L., Minaei, M., et al., 2023. Stable isotope signatures in tehran's precipitation: insights from artificial neural networks, stepwise regression, wavelet coherence, and ensemble machine learning approaches. Water. https://doi.org/10.3390/w15132357.

Hu, H., Wang, J.L., 2015. Characteristics of hydrogen and oxygen isotopes in atmospheric precipitation and analysis of moisture sources during October-December 2013 in Chongqing. Carsol. Sinica 34 (3), 253–260. https://doi.org/10.11932/karst20150307.

IAEA/WMO, 1992. Global Network of Isotopes in Precipitation, GNIP database. https://www.iaea.org/water.

Jin, Z., Wang, Y., Li, F., Qian, L., Hu, Y., Shi, Y., 2019. Stable isotopes and chemical characteristics of precipitation in Hangzhou and Huzhou, East China. Environ. Sci. Pollut. Res. 26 (23), 23717–23729. https://doi.org/10.1007/s11356-019-05712-y.

Kalnay, E.M., Kanamitsu, M., Kistler, R., et al., 1996. The Nmc/ncar 40-year reanalysis project. Bull. Am. Meteorol. Soc. 77 (3). https://doi.org/10.1175/1520-0477(1996)0772.0.CO;2.

Kralik, M., Papesch, W., Stichler, W., 2003. Austrian Network of Isotopes in Precipitation (ANIP): Quality Assurance and Climatological Phenomenon in One of the Oldest and Densest Networks in the World. Isotope Hydrology and Integrated Water Resources Management International Symposium Held in Vienna, Austria. 146–149.

Kuhn. et al., 2020. Tidymodels: a collection of packages for modeling and machine learning using tidyverse principles. https://www.tidymodels.org.

Li, J., Pang, Z., Tian, L., Zhao, H., Bai, G., 2022. Variations of stable isotopes in daily precipitation in a monsoon region. Water 14 (18), 2891. https://doi.org/10.3390/w14182891.

Liaw, A., Wiener, M., 2002. Classification and regression by randomForest. *R News 2 (3), 18–22.

Liu, J., et al., 2008. Characteristics of δ^{18}O in precipitation over northwest China and its water vapor sources. Acta Geograph. Sin. 12–22.

Liu, J., Song, X., Yuan, G., Sun, X., Liu, X., Wang, S., 2010. Characteristics of δ 18O in precipitation over Eastern Monsoon China and the water vapor sources. Chin. Sci. Bull. 55 (2), 200–211. https://doi.org/10.1007/s11434-009-0202-7.

Liu, J., Song, X., Yuan, G., Sun, X., Yang, L., 2014. Stable isotopic compositions of precipitation in China. Tellus B Chem. Phys. Meteorol. 66 (1), 22567. https://doi.org/10.3402/tellusb.v66.22567.

Liu, Z.F., Tian, L.D., Yao, T.D., Chai, X.R., 2009. Spatial distribution of δ^{18}O in precipitation over China. Chin. Sci. Bull. 54 (6), 8. https://doi.org/10.1360/csb2009-54-6-804.

Lynch, J.A., et al., 1995. Trends in precipitation chemistry in the United States: a National Perspective, 1980–1992. Atmos. Environ. 29, 1231–1246. https://doi.org/10.1016/1352-2310(94)00371-Q.

Meng, Y., Liu, G., 2010. Effect of below-cloud secondary evaporation on the stable isotopes in precipitation over the Yangtze River basin. Chinese Adv. Water Sci. 21 (3), 327–334. https://doi.org/10.1541/ieejpes.130.61.

Nelson, D.B., Basler, D., Kahmen, A., 2021. Precipitation isotope time series predictions from machine learning applied in Europe. Proc. Natl. Acad. Sci. 118 (26), 2024107118. https://doi.org/10.1073/pnas.

Rao, W.B., Li, Y.W., Tan, H.B., Li, Y.G., Wen, C., Zhang, X.Y., 2021. Hydrogen and oxygen stable isotopic composition of precipitation and its moisture sources in an alpine arid region: a case study of the Golmud River Basin on the northern slope of Kunlun Mountains. J. Hydraul. Eng. 52 (9), 1116–1125. https://doi.org/10.13243/j.cnki.slxb.20201087.

Rhoujjati, N., Ait Brahim, Y., Hanich, L., Rhoujjati, A., Patris, N., Chehbouni, A., Bouchaou, L., 2023. Precipitation isotopes to elucidate moisture sources in the Western Mediterranean: case of the Middle Atlas Mountains, Morocco. Environ. Earth Sci. 82 (10). https://doi.org/10.1007/s12665-023-10930-2.

Shi, X., Risi, C., Pu, T., Lacour, J.L., Kong, Y., Wang, K., He, Y.Q., Xia, D.S., 2020. Variability of isotope composition of precipitation in the Southeastern Tibetan Plateau from the synoptic to seasonal time scale. J. Geophys. Res. Atmos. 125 (6), e2019JD031751. https://doi.org/10.1029/2019JD031751.

Shi, Y., Wu, A., Li, G., Li, F., 2021. Stable isotopic characteristics of precipitation related to the environmental controlling factors in Ningbo, East China. Environ. Sci. Pollut. Res. 28 (9), 10696–10706. https://doi.org/10.1007/s11356-020-11332-8.

Song, X.F., Liu, J.R., Sun, X.M., et al., 2007. Establishment of Chinese network of isotopes in precipitation (CHNIP) based on CERN. Adv. Earth Science 22 (7). DOI: 10.10.3321/j.issn:1001-8166.2007.07.010.

Soetaert, K., 2024. shape: Functions for Plotting Graphical Shapes, Colors. R package version 1.4.6.1, <https://CRAN.R-project.org/package=shape>.

Terzer, S., Wassenaar, L.I., Araguás-araguás, L.J., et al., 2013. Global isoscapes for δ^{18}o and δ^{2}H in precipitation: improved prediction using regionalized climatic regression models. Hydrol. Earth Syst. Sci. 17 (11), 4713–4728. https://doi.org/10.5194/hess-17-4713-2013.

Tian, L., Yao, T., MacClune, K., White, J.W.C., Schilla, A., Vaughn, B., Ichiyanagi, K., 2007. Stable isotopic variations in west China: a consideration of moisture sources. J. Geophys. Res. 112, D10112. https://doi.org/10.1029/2006JD007718.

Van, den. Brand. T., 2024. ggh4x: Hacks for 'ggplot2'. R package version 0.2.8. https://CRAN.R-project.org/package=ggh4x>.

Wallace, J.M., Gutzler, D.S., 1981. Teleconnections in the geopotential height field during the Northern Hemisphere winter. Weather Rev. 109, 784–812. https://doi.org/10.1175/1520-0493(1981)109%3C0784:TTTGHF%3E2.0.CO;2.

Wang, H.J., Zhang, J.L., Liu, Z.H., 2012. Climatic implications of hydrogen and oxygen isotopes in precipitation from Huanglong, Sichuan Province. Carsol. Sinica 31 (3), 293–298.

Wang, L., Dong, Y., Han, D., Xu, Z., 2019. Stable isotopic compositions in precipitation over wet island in Central Asia. J. Hydrol. 573, 581–591. https://doi.org/10.1016/j.jhydrol.2019.04.005.

Wang, Q., Wang, Y., Zhao, K., Chen, S., Liu, D., Zhang, Z., Liang, Y., 2018. The transfer of oxygen isotopic signals from precipitation to drip water and modern calcite on the seasonal time scale in Yongxing Cave, central China. Environ. Earth Sci. 77 (12), 474. https://doi.org/10.1007/s12665-021-09504-x.

Wang, S., Lei, S., Zhang, M., Hughes, C., Crawford, J., Liu, Z., Qu, D., 2022a. Spatial and seasonal isotope variability in precipitation across China: monthly isoscapes based on regionalized fuzzy clustering. J. Clim. 35 (11), 3411–3425. https://doi.org/10.1175/JCLI-D-21-0451.1.

Wang, S.J., Wang, L.W., Zhang, M.J., 2023. Methodology and application of precipitation isoscapes of stable hydrogen and oxygen isotopes. Chin. J. Appl. Ecol. 34 (2), 566–576. https://doi.org/10.13287/5j.1001-9332.202302.016.

Wang, T., et al., 2022b. Spatial and temporal distributions of precipitation isotope information entropy to trace water vapor transport in China. Chin. Adv. Water Sci. 33 (4), 11. https://doi.org/10.14042/j.cnki.32.1309.2022.04.006.

Wang, X., Li, Z., Tayier, R., Wang, S., 2015. Characteristics of atmospheric precipitation isotopes and isotopic evidence for the moisture origin in Yushugou River basin, Eastern Tianshan Mountains, China. Quat. Int. 380, 106–115. https://doi.org/10.1016/j.quaint.2014.12.023.

Wu, H., Zhang, X., Li, X., et al., 2015. Seasonal variations of deuterium and oxygen-18 isotopes and their response to moisture source for precipitation events in the subtropical monsoon region. Hydrol. Process. 29 (1), 90–102. https://doi.org/10.1002/hyp.10132.

Wu, X., Zhu, X.Y., Zhang, M.L., Bai, X., Zhang, B.Y., 2013. High-resolution record of stable isotopic composition in atmospheric precipitation: a case study of Guilin area. Resour. Environ. Yangtze Basin 22 (2), 182–188.

Xing, G., Guofeng, Z., Dan, D., Han, X.P., Xiu, T.X., Rui, F.Y., 2018. Stable isotope composition of precipitation at different elevations in the monsoon marginal zone. Quat. Int. 493, 86–95. https://doi.org/10.1016/j.quaint.2018.06.038.

Yang, Y., 2020. Characteristics and environmental significance of hydrogen and oxygen isotopes in atmospheric water vapor and precipitation in Shanghai [Doctoral dissertation, East China Normal University]. DOI: 10.27149/d.cnki.ghdsu.2020.000252.

Yao, T., Masson-Delmotte, V., Gao, J., Yu, W., Yang, X., Risi, C., Hou, S., 2013. A review of climatic controls on δ^{18}O in precipitation over the Tibetan Plateau: Observations and simulations. Rev. Geophys. 51 (4), 525–548. https://doi.org/10.1002/rog.20023.

Yu, W., et al., 2016. Stable isotope variations in precipitation over Deqin on the southeastern margin of the Tibetan Plateau during different seasons related to various meteorological factors and moisture sources. Atmos. Res. 170, 123–130. https://doi.org/10.1016/j.atmosres.2015.11.013.

Zeng, K.K., Yang, Y.H., Hu, Y.C., et al., 2021. Characteristics of precipitation isotopes and analysis of moisture sources in the Kashi River Basin. Arid Zone Res. 38 (05), 1263–1273. https://doi.org/10.13866/j.azr.2021.05.08.

Zhan, Z., Pang, H., Wu, S., Liu, Z., Zhang, W., Xu, T., Hou, S., 2023. Determining key upstream convection and rainout zones affecting δ18O in water vapor and precipitation based on 10-year continuous observations in the East Asian Monsoon region. Earth Planet. Sci. Lett. 601, 117912. https://doi.org/10.1016/j.epsl.2022.117912.

Zhang, B.X., 2020. Characteristics of Stable Isotopes in Atmospheric Precipitation and Moisture sources in Shijiazhuang. Hebei Normal University, 10.27110/d.cnki.ghsfu.2020.001334.

Zhang, F., Huang, T., Man, W., Hu, H., Long, Y., Li, Z., Pang, Z., 2021. Contribution of recycled moisture to precipitation: a modified D-excess-based model. Geophys. Res. Lett. 48 (21), e2021GL095909. https://doi.org/10.1029/2021GL095909.

Zhang, G.L., et al., 2015. Hydrogen and oxygen isotopes in precipitation in Southwest China: progress and prospects. Chinese J. Glaciol. Geocryol. 37 (4), 10. https://doi.org/10.7522/j.issn.1000-0240.2015.0122.

Zhang, X., Sun, C.J., Zhou, S.J., Wang, S.Y., Chen, W., 2023. Assessment of spatial and seasonal variability on multi-scale precipitation isotopes in the high mountains in Western China. Chinese Geogr. Res. 42 (5), 1393–1409. https://doi.org/10.11821/dlyj020220736.

Zhang, X.P., Yao, T.D., 1994. Relations of oxygen isotopic composition in precipitation with temperature and precipitation amount in some regions of China. Chinese J. Glaciol. Geocryol. 016 (1), 31–40.

Zhang, X., Jin, H.J., Sun, W., 2006. Stable isotopic variations in precipitation in Southwest China. Adv. Atmos. Sci. 23 (4), 649–658. https://doi.org/10.1007/s00376-006-0649-2.

Zhao, J.C., Wei, B.H., Xiao, S.B., 2009. Characteristics of stable isotopes in atmospheric precipitation in Yichang area, Hubei Province. Tropical Geogr. 29 (6), 526–531.

Zhao, P., Tan, L., Zhang, P., Wang, S., Cui, B., Li, D., Xue, G., Cheng, X., 2018. Stable isotopic characteristics and influencing factors in precipitation in the monsoon marginal region of northern China. Atmos. 9 (3), 97. https://doi.org/10.3390/atmos9030097.

Zhao, S., Pang, S., Wen, R., et al., 2015. Influence of below-cloud secondary evaporation on stable isotope composition in precipitation in the Haihe River Basin, China. Chinese Progr. Geogr. 34 (8), 1031–1038. https://doi.org/10.18306/dlkxjz.2015.08.010.

Zhao, W., 2017. Characteristics of precipitation isotopes and moisture sources in the Shule River Basin [Doctoral dissertation, Lanzhou University, Lanzhou, China].

Zhu, G., Liu, Y., Shi, P., Jia, W., Zhou, J., Liu, Y., Zhao, K., 2022a. Stable water isotope monitoring network of different water bodies in Shiyang River basin, a typical arid river in China. Earth Syst. Sci. Data Discuss. 2022, 1–23. https://doi.org/10.5194/essd-2021-79.

Zhu, X., Xiao, W., Wang, J.Y., et al., 2022b. Characteristics of hydrogen and oxygen isotopes in precipitation in Nanjing. J. Appl. Meteorol. Sci. 33 (03), 353–363. https://doi.org/10.11898/1001-7313.20220309.